Q1.

(a) Pineapple juice contains a substance that speeds up the digestion of protein.

(i) What is the name for substances that speed up digestion?

.....

1 mark

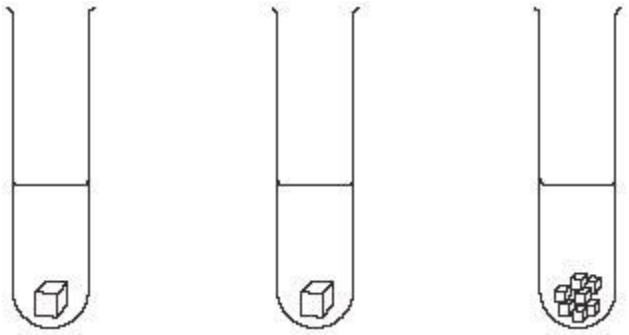
(ii) What happens to a molecule of protein during digestion?

.....

.....

1 mark

(b) Asim did an experiment to investigate the digestion of gelatin.
Gelatin is the protein in jelly.
In test tubes A and B he used one cube of jelly in each.
In test tube C he used one cube of jelly that he had chopped up.



A
5 g jelly cube
+
15 cm^3
cold water

B
5 g jelly cube
+
15 cm^3 fresh
pineapple
juice

C
5 g jelly cube
chopped into
pieces
+
15 cm^3 fresh
pineapple
juice

He recorded how long it took for the jelly to be digested in each test tube.
The table below shows his results.

test tube	result
A	**not** digested after 2 hours
B	jelly digested in 2 hours
C	jelly digested in 1 hour

(i) What was the purpose of test tube A?

.....

.....

1 mark

(ii) It is helpful to chew your food.
How do the results in test tube C show this?

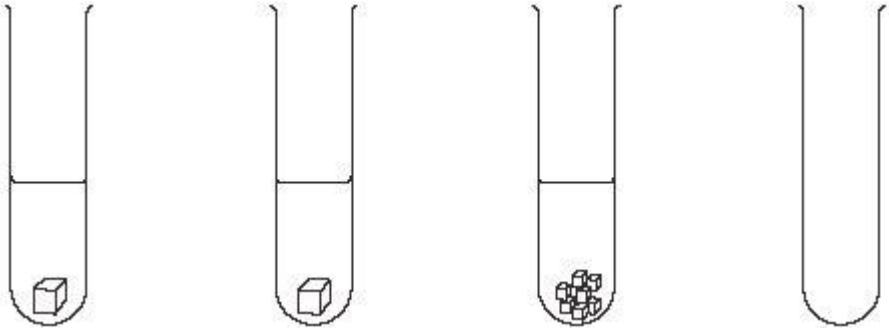
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1 mark

(c) The substances that speed up digestion stop working when they have been boiled.

(i) What does Asim need to put in a fourth test tube to test this in his experiment?
Label test tube D.
Test tubes A, B and C contain the same as in the first experiment.



A	**B**	**C**	**D**
5 g jelly cube + 15 cm^3 cold water	5 g jelly cube + 15 cm^3 fresh pineapple juice	5 g jelly cube chopped into pieces + 15 cm^3 fresh pineapple	.... + ...

2 marks

(ii) Predict what Asim would observe in test tube D after 2 hours.

.....

1 mark
maximum 7 marks

Q1.

(a) (i) enzyme(s)

accept '(hydrochloric) acid'
*do **not** accept other named acids*

accept 'biological catalyst'
'catalyst' is insufficient

*accept any named enzyme (e.g. amylase **or** protease)*
'saliva' is insufficient

1 (L7)

(ii) it is broken down (into smaller molecules)

accept 'forms amino acids'
'it is absorbed' is insufficient

accept 'breaks up'
*do **not** accept 'it breaks down into glucose'*

1 (L6)

(b) (i) any **one** from

- as a control

  accept 'control'
  'so that it is a fair test' is insufficient

  accept 'as a comparison'

- to show that enzyme digested the jelly

  accept 'to show what would happen without the enzyme or pineapple'

- to show that water does not digest the jelly

1 (L7)

(ii) any **one** from

- it took less time to digest the jelly (than B)

  *accept 'it breaks it down **or** up more quickly'*
  an explanation of why it is faster is insufficient

- chopped up jelly digested more quickly

  'it is easier to swallow' is insufficient
  'it is easier to digest' is insufficient

  *accept 'it **only** takes an hour to digest'*
  'it digests in an hour' is insufficient

1 (L6)

(c) (i) • 5 g cube of (chopped up) jelly **and** same amount **or** 15 cm^3 of juice

accept 'use same amount of jelly and juice'
*do **not** accept 'water'*

1 (L7)

- boil the juice first

  accept 'use (fresh) boiled juice'

*do **not** accept 'boiling juice'*

1 (L6)

(ii) the jelly was not digested

accept 'how much jelly had broken down'

accept 'nothing happened'
'nothing' is insufficient as it implies nothing is left in the test tube

accept '5 g of jelly (cubes)'

accept 'a small amount of jelly has dissolved'
'the jelly has not dissolved' is insufficient

1 (L6)

[7]